From Variants to Structure: Assessing Protein Structures of Hypertension-Associated Genes Using AlphaFold3 and TM-score/RMSD

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August 28, 2025

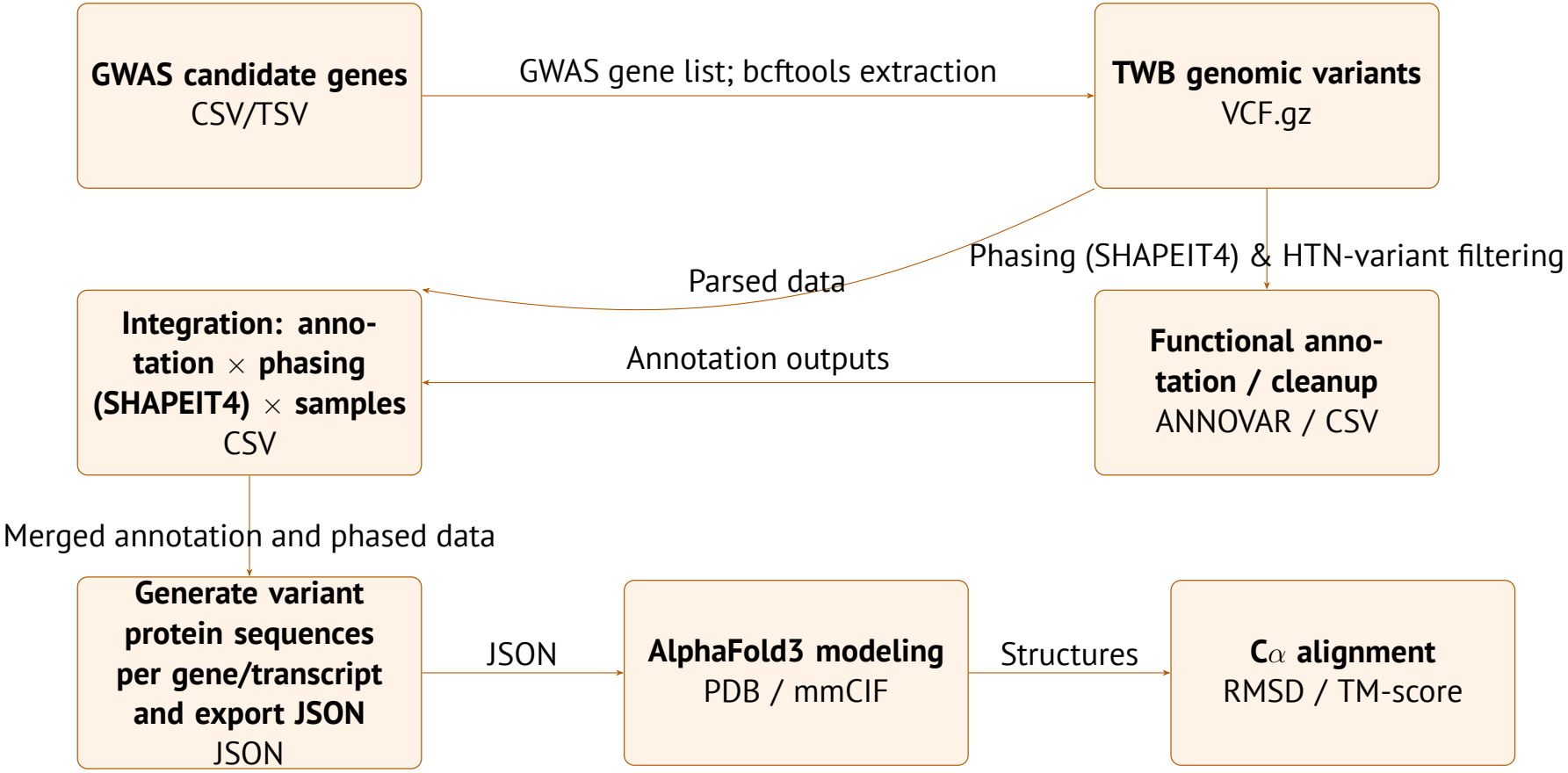
Motivation

- Research Question: Do hypertension-associated genetic variants actually alter protein structures?
- Research Objective: To examine whether structural differences are associated with participants' hypertension status.

Data Sources

1. **GWAS Catalog**
   - A public database maintained by EMBL-EBI and the U.S. NHGRI.
   - Curators manually read and verify published GWAS studies, standardize them into a unified format, and compile **SNP–trait associations**.
   - In this study, candidate genes associated with hypertension were retrieved from this database.
2. **Ensembl**
   - A genome information platform maintained by EMBL-EBI and the Wellcome Sanger Institute.
   - Used in this study to obtain the start and end coordinates of candidate genes and convert them into standardized genomic coordinates (chr, start, end).
3. **Taiwan Biobank (TWB)**
   - A nationwide cohort and biobank project in Taiwan.
   - This study used whole-genome variant data (VCF) provided by TWB, along with **1,491** participants' questionnaire and physical examination data.

Workflow: From Variants to Structure



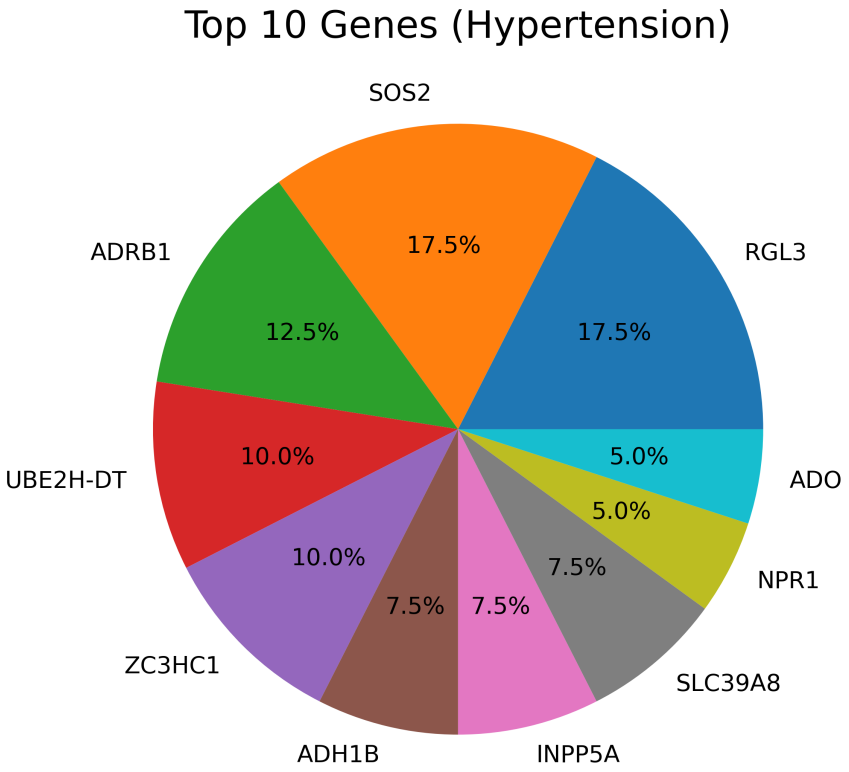
From Variants to Structure *Workflow Diagram*

References:

Abramson, J., AlQuraishi, M., Baker, D., *et al.* (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630, 493–500. https://doi.org/10.1038/s41586-024-07487-w

Selecting Hypertension Associated Genes: *RGL3*, *NPR1*, *ADRB1*

1. **Disease keywords (strictly HTN):**
   - treatment-resistant hypertension
   - early-onset hypertension
   - pulmonary arterial hypertension
   - essential hypertension
   - hypertension
2. **Significance threshold:**
   - *P*-value $< 5 \times 10^{-8}$
3. **Filtering criterion:**
   - `CONTEXT` is missense variant



Top 10 Genes (Hypertension)

Structural Alignment Methods

- Global alignment: Cα atoms (whole protein)
- Local alignment: Functional regions / domains
- Evaluation metrics: TM-score and RMSD

$$\text{RMSD} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\|p_i - q_i\|^2}$$

$$\text{TM-score} = \frac{1}{L_{\text{target}}}\sum_{i=1}^{L_{\text{align}}}\frac{1}{1+\left(\frac{d_i}{d_0(L_{\text{target}})}\right)^2}$$

- N: number of aligned atoms
- p_i, q_i: coordinates of atom i after optimal rotation/translation
- $d_i = \|x'_i - y'_i\|$
- $d_0(L) = \max\left(0.5, 1.24(L-15)^{1/3} - 1.8\right)$
- x'_i: atom i in the aligned structure
- y'_i: atom i in the reference structure
- L_{target}: length of the target structure (residues)
- L_{align}: number of residues actually aligned

References:

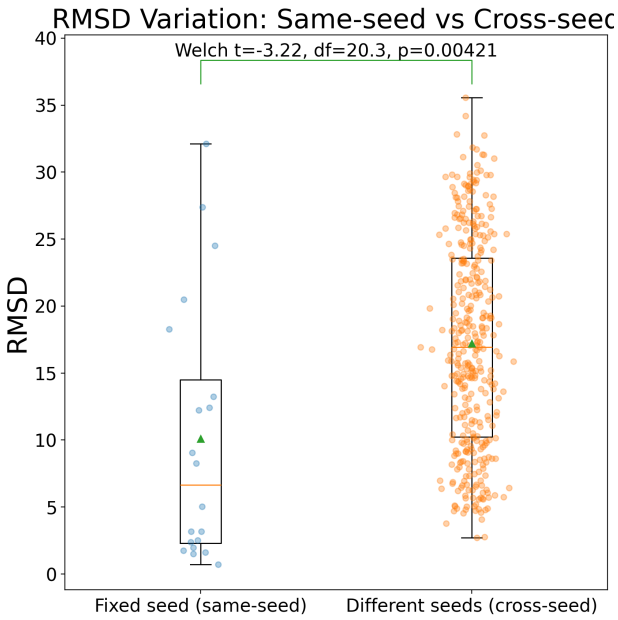
Kabsch, W. (1976). A solution for the best rotation to relate two sets of vectors. *Acta Crystallographica Section A*, 32(5), 922–923. (RMSD)

Zhang, Y., & Skolnick, J. (2004). Scoring function for automated assessment of protein structure template quality. *Proteins*, 57(4), 702–710. (TM-score)

Tested the effect of using a fixed seed vs. random seeds on the RMSD between reference and mutated protein sequences.

- Tested the effect of using a fixed seed vs. random seeds.
- Result: Fixed seed produced more stable structural alignments.

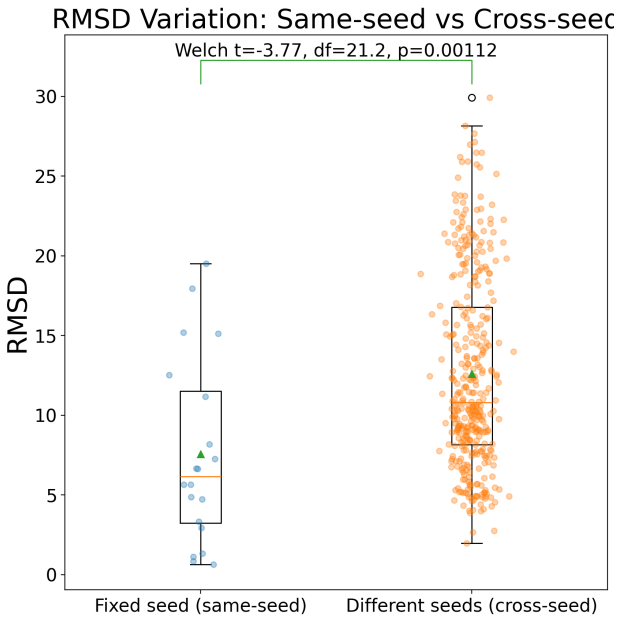
$p = 0.00421$



NPR1

Fixed seed vs. Random seed

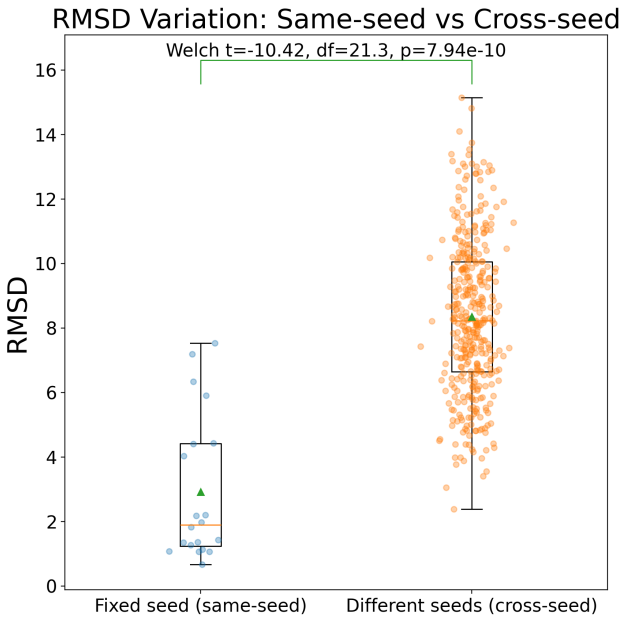
$p = 0.00112$



RGL3

Fixed seed vs. Random seed

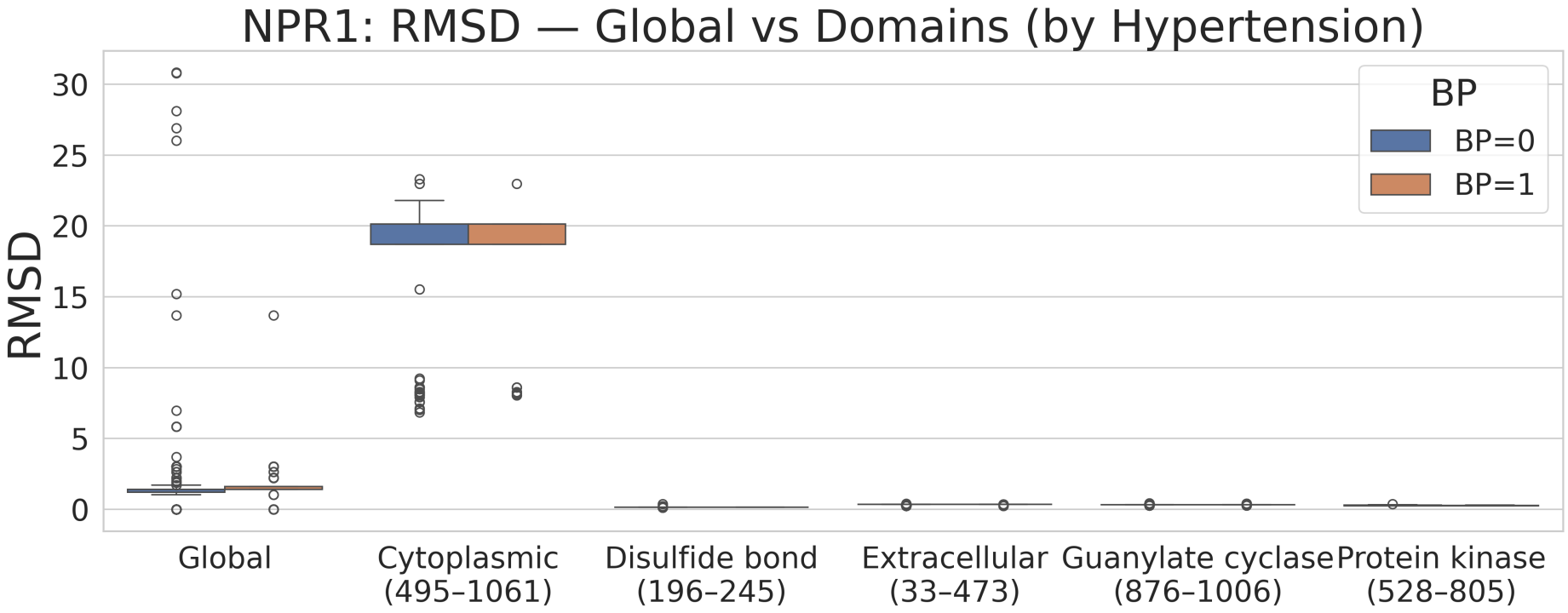
$p = 7.94e - 10$



ADRB1

Fixed seed vs. Random seed

Structural Differences in *NPR1* but No Association With Hypertension

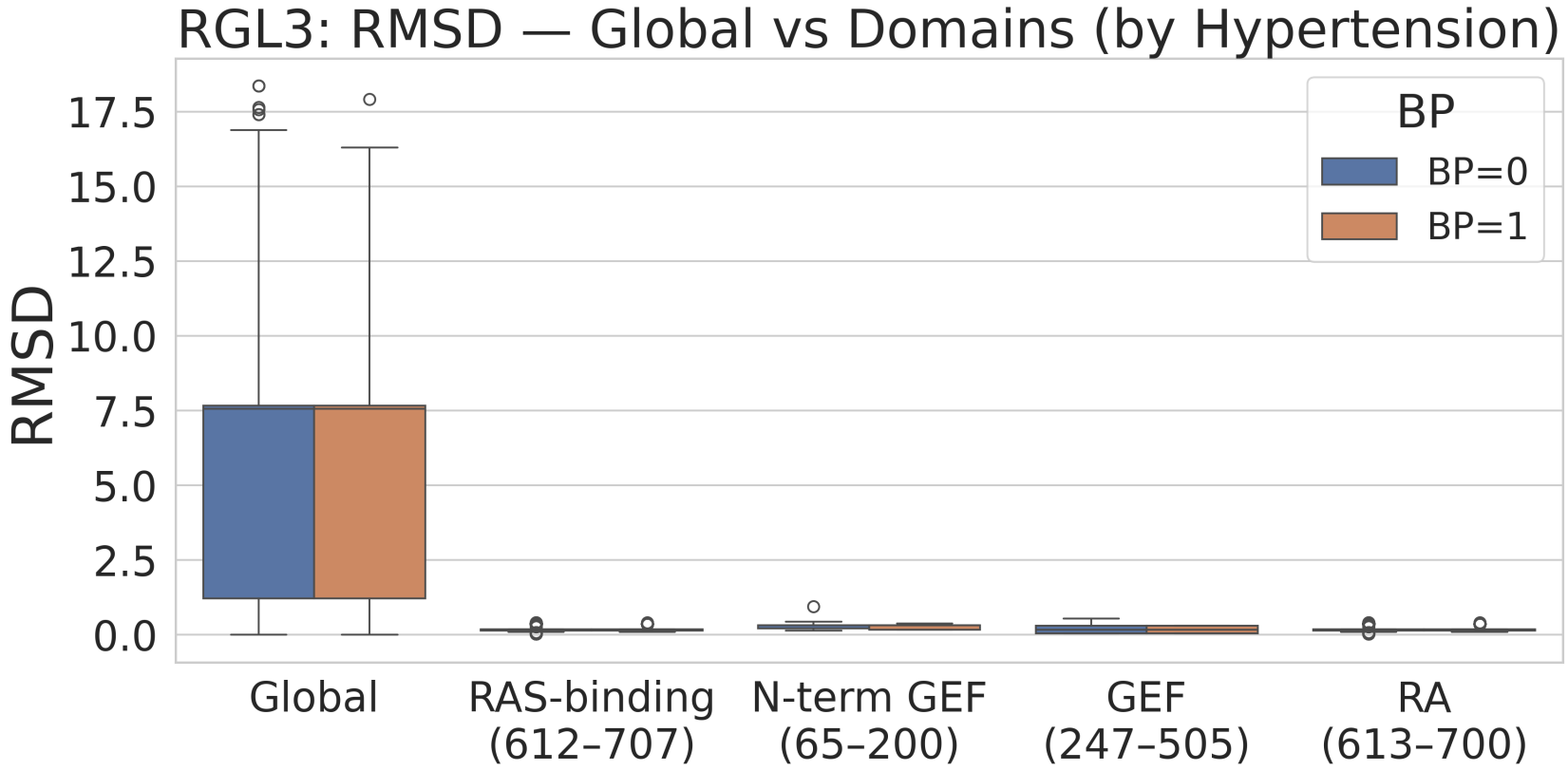


NPR1

Global vs Domains (by Hypertension)

No significant difference by hypertension status under either global or domain-level alignment.

Structural Differences in *RGL3* but No Association With Hypertension



RGL3

Global vs Domains (by Hypertension)

No significant difference by hypertension status under either global or domain-level alignment.

Conclusions and Future Work

Conclusions

- Built a reproducible end-to-end pipeline: starting from GWAS genes and TWB VCFs, generated H1/H2 haplotype variant protein sequences and modeled them with AlphaFold3.
- Focused on missense variants and selected three target genes: *RGL3*, *ADRB1*, and *NPR1*.
- Results showed that fixing the seed produced more stable models.
- Structural differences were observed in *RGL3* and *NPR1*, but these did not clearly separate hypertension groups.
- At the domain level, only the cytoplasmic region of *NPR1* showed a shift, but it was not significantly associated with hypertension.

Future Work

- Extend the analysis to more genes and samples to increase statistical power.
- Explore additional structural alignment and functional prediction metrics to complement RMSD and TM-score.

Acknowledgements

We thank the Institute of Statistical Science Academia Sinica and Smart Health Project –the Statistical Science unit for the facility and financial support.

Thank you.

Tools for Data Processing and Modeling

- **bcftools** (v1.19): Processing and filtering VCF files; extracting candidate gene regions.
- **SHAPEIT4** (v4.2.0): Genotype phasing; converts 0/1 into 0|1 and generates H1/H2 haplotypes.
- **ANNOVAR** (2020Jun): Functional annotation tool; annotates gene and protein variants.
- **AlphaFold3** (2024): Protein structure modeling; predicts 3D structures of variant sequences.
- **PyMOL** (v3.1.0, Open-Source, 2025-05-02): Used for protein structure superposition and alignment; computes RMSD and TM-score.

References:

Danecek, P., Bonfield, J. K., Liddle, J., *et al.* (2021). Twelve years of SAMtools and BCFtools. *GigaScience*, 10(2), giab008. doi:10.1093/gigascience/giab008

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Wang, K., Li, M., & Hakonarson, H. (2010). ANNOVAR: functional annotation of genetic variants from high-throughput sequencing data. *Nucleic Acids Research*, 38(16), e164. doi:10.1093/nar/gkq603

Abramson, J., AlQuraishi, M., Baker, D., *et al.* (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630, 493–500. doi:10.1038/s41586-024-07487-w

DeLano, W. L. (2002). The PyMOL Molecular Graphics System. DeLano Scientific. http://www.pymol.org

VCF File Format (GRCh38/hg38)

Column	Description
CHROM	Chromosome
Start	Start position
End	End position
REF	Reference allele
ALT	Alternative allele
FORMAT	Sample format definition
SAMPLE	Genotype and related information per sample

CHROM	Start	End	REF	ALT	FORMAT	SAMPLE1	SAMPLE2
chr1	153679113	153679113	C	T	GT	0/1	0/0

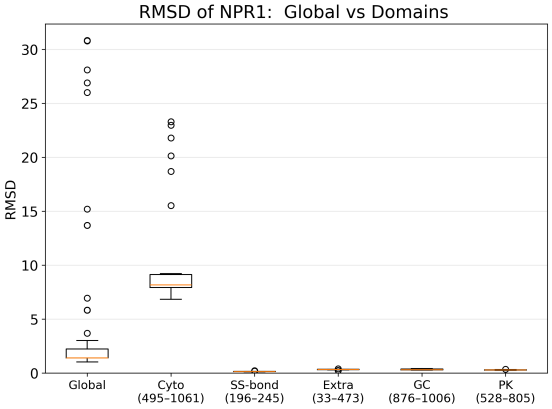
Target Gene introduction

Gene	Transcript	Chromosome	Length (aa)	Association with Hypertension
RGL3	NM_001035223	chr19	710	Implicated in BP regulation via GWAS findings
ADRB1	NM_000684	chr10	477	Beta-1 adrenergic receptor; variants affect BP and cardiac output
NPR1	NM_000906	chr1	1061	Natriuretic peptide receptor 1; regulates sodium excretion and blood pressure

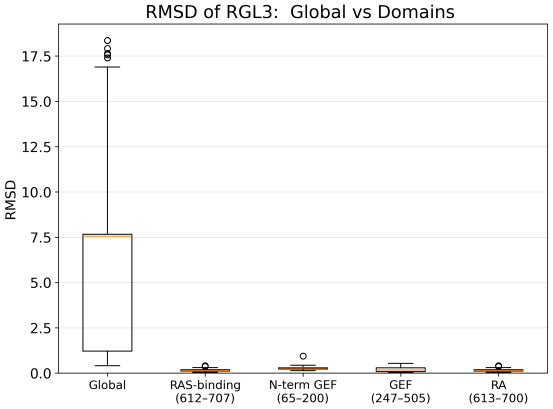
Note: Length refers to protein sequence length in amino acids (aa). "Association with Hypertension"summaries are based on GWAS findings and known physiological roles.

Structural Alignment Results (Global vs. Domains)

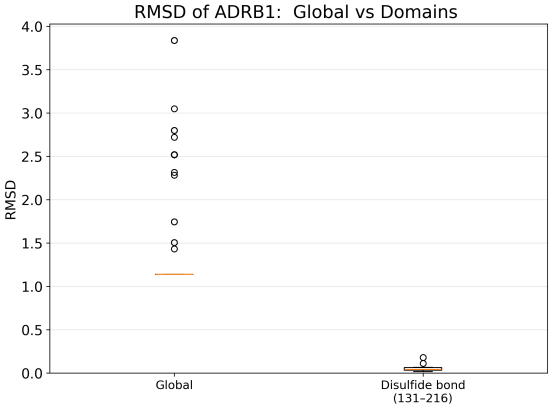
- Performed both global and local structural alignments for RGL3, ADRB1, and NPR1.
- Calculated RMSD and TM-score for evaluation.



NPR1 Global vs Domains



RGL3 Global vs Domains



ADRB1 Global vs Domains